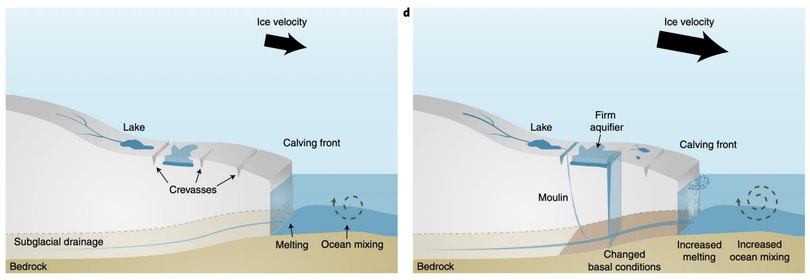
Satellite and field observations of the surface hydrology and ice dynamics of Antarctic Peninsula glaciers

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Introduction

- Surface melting is widespread in the Antarctic Peninsula and will likely increase as the atmosphere warms (Davies et al., 2025).
- Surface melting can affect the flow of outlet glaciers where it reaches the bed, frequently via lake drainage (Stevens et al., 2022).
- Whether meltwater affects the flow of Antarctic Peninsula outlet glaciers is controversial (Tuckett et al., 2019; Rott et al, 2019; Tuckett et al, 2020).

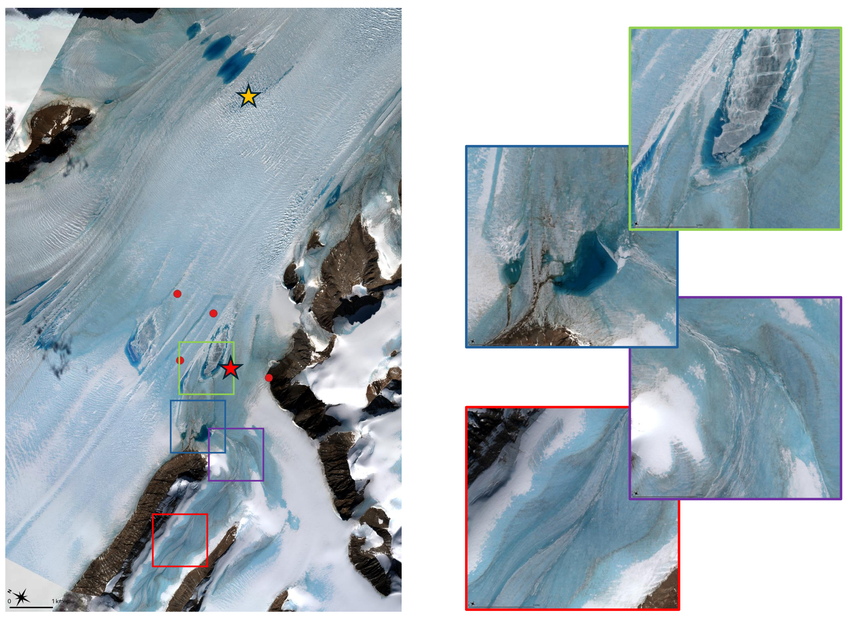


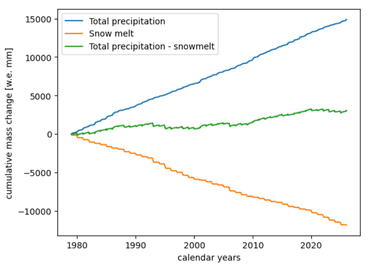
Cartoon from Bell et al. (2018) depicting the effect of surface melting water on ice dynamics through modification of basal conditions.

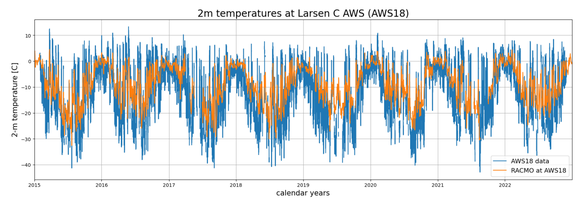
The Flow Response of Antarctic ice to Meltwater (FRAM) project is using remote sensing and field obersvations to test whether meltwater reaches the bed of Antarctic Peninsula glaciers and modulates their flow.

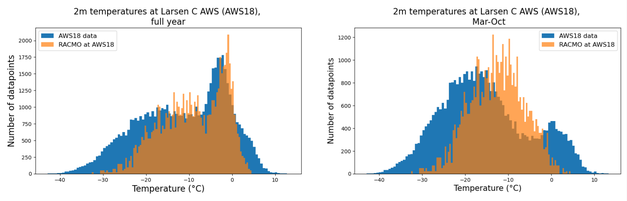
Surface conditions on Flask Glacier:

Remote sensing and field observations reveal large firn-free regions, an extensive surface drainage system, and multiple connected supraglacial lakes. Modelled accumulation slightly exceeds modelled melt, but comparison between modelled and measured air temperatures indicate melt may be underestimated.





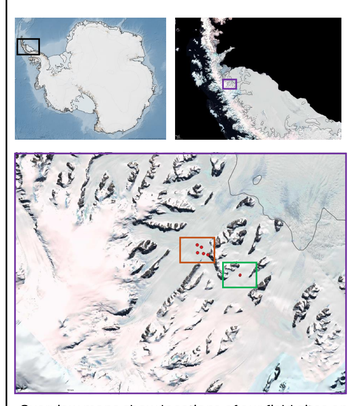




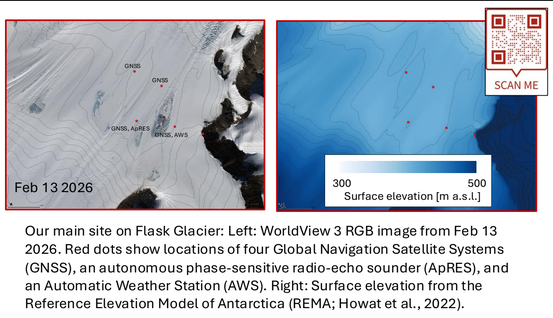
WorldView 2 RGB image of Flask Glacier from Jan 2 2025. Red dots show our instrument locations. Insets show closer views of surface hydrological features. Inset locations are shown by the corresponding coloured squares in the main panel. Stars indicate locations of UAV and GNSS data displayed in the lower-left panel of this poster.

Measured and modelled meteorology on Flask. Top: Two cumulative surface-mass-balance components and their difference at our field site according to 2km-downscaled- RACMO2.3p2 interpolated at the AWS (Noël et al. 2013; *pers comm. B. Noël*). Middle: Time series of 2-m air temperature from an AWS on Larsen C Ice Shelf, 75 km south of Flask (AWS18; van Tiggelen et al., 2025) and RACMO interpolated at the AWS. Bottom: Histograms of 2-m air temperature according RACMO and AWS18 (2015-2023). Left histogram shows year-round values. Right histogram shows values for March to October.

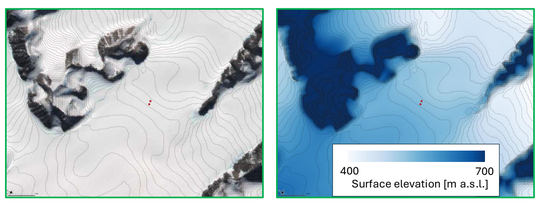
Flask Glacier



Overview maps show locations of our field sites on the Antarctica Peninsula. Red and green squares show locations of closer views to the right.

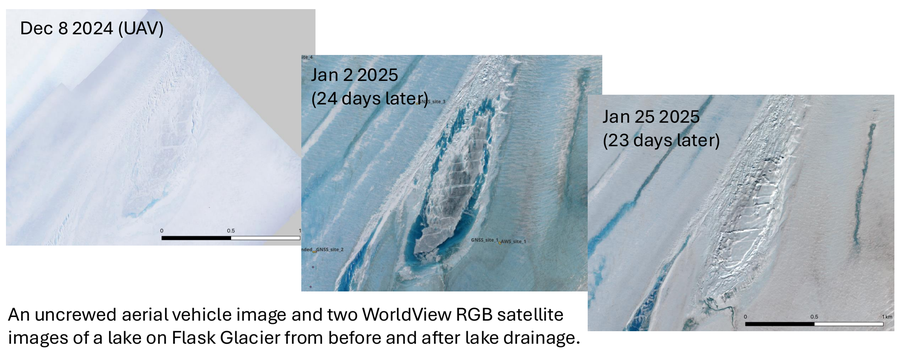


Our main site on Flask Glacier: Left: WorldView 3 RGB image from Feb 13 2026. Red dots show locations of four Global Navigation Satellite Systems (GNSS), an autonomous phase-sensitive radio-echo sounder (ApRES), and an Automatic Weather Station (AWS). Right: Surface elevation from the Reference Elevation Model of Antarctica (REMA; Howat et al., 2022).

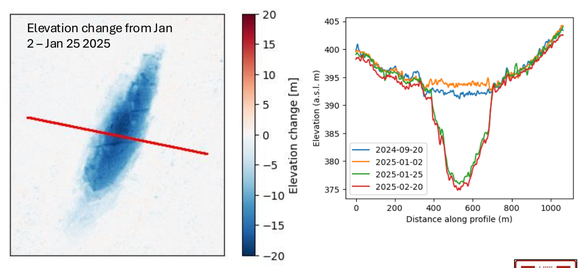


Our secondary site on a platuea adjacent to Flask Glacier: Left: Landsat Image Mosaic of Antarctica Red dots indicate locations a GNSS, an AWS, an ApRES, and a passive seismometer. Right: Surface elevation from REMA.

Lake drainage: Our GNSS, ApRES and AWS were deployed around a 1.5-km long, 600-m wide lake that drained in Jan 2025 leaving a 2,500,000 m^3 depression.



An uncrewed aerial vehicle image and two WorldView RGB satellite images of a lake on Flask Glacier from before and after lake drainage.



The images shown to the left and two additional WorldView images were used to create digital elevation models from which a difference map (left) and multiple elevation profiles (right) were produced.

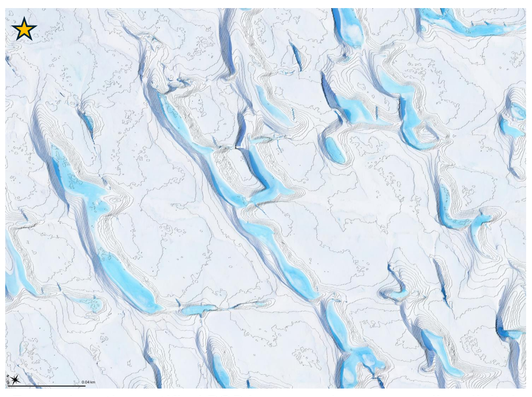
Fieldwork and preliminary data: two field teams deployed an instrument array on Flask and an adjacent higher-elevation region to the south. Access challenges have delayed data retrieval until November 2026.



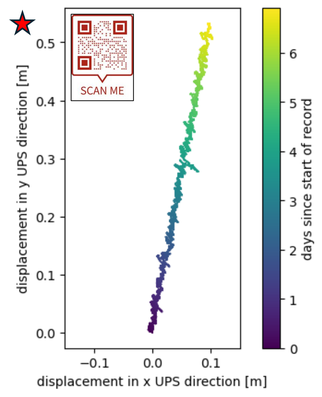
The field team on Flask Glacier, Dec 2024



Instrument deployments on Flask. Left two panels: K. Winter with ApRES and GNSS installations. Right two panels: R. Muthyala with an AWS installation and deploying a WingtraOne Gen II uncrewed aerial vehicle.



Example orthorectified RGB image and corresponding digital elevation model collected using a WingtraOne Gen II uncrewed aerial vehicle on Flask. See image in 'Surface conditions' section for location. 50 cm contours are shown in black.

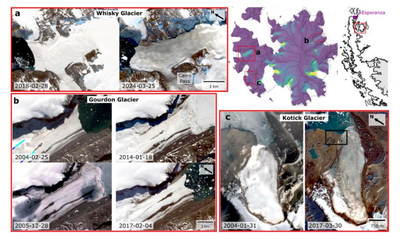


GNSS displacement data over one week following installation at Site 1 in December 2024 (location shown in 'Surface conditions' section).

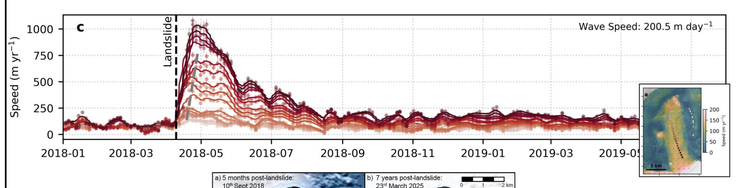
Remotely sensed ice-surface velocities: Peninsula-wide ice velocity fields were derived using intensity tracking algorithms (SHeffield Ice Flow Tracker, SHIFT) applied to consecutive Sentinel-1 image pairs.

Velocity fields have been used to explore temporal variability in velocity related to...

...surging...

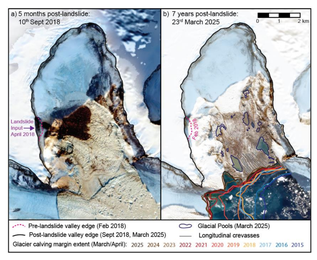


...supgraglacial landslides (Davison et al., in prep.)...

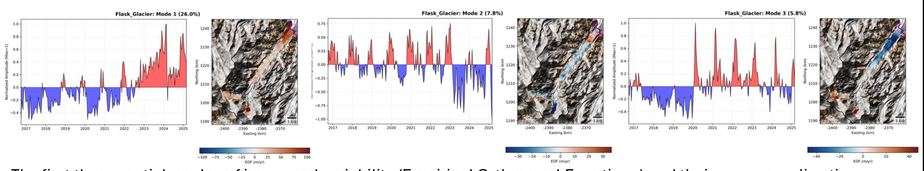


Surging glaciers on JRI. Left: Before- and after-surge imagery of Whisky, Gourdon, and Kotick Glaciers. Right: Velocity maps and time series capturing the surge on Whisky Glacier (Davison et al., in press).

...and ocean or atmospheric variability.



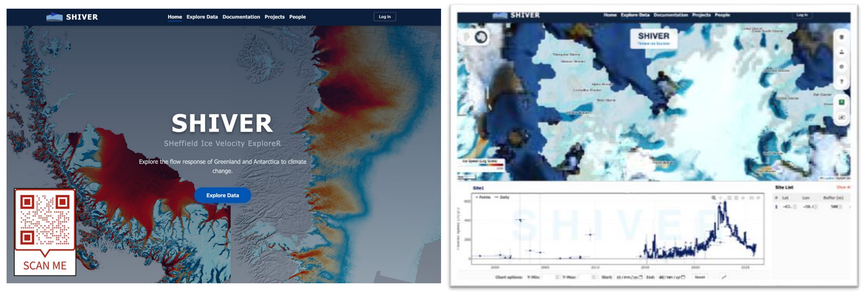
Right: Sentinel 2 RGB imagery from immediately after and 7 years after a supraglacial landslide on Swift Glacier. Top: ice speed time series indicating a velocity wave in the months following the landslide.



The first three spatial modes of ice-speed variability (Empirical Orthogonal Functions) and their corresponding time series, for Flask Glacier. Maps are relative to 2016-2025 average speed.

SHeffield Ice Velocity ExploreR (SHIVER):

A new portal for exploring and downloading velocity data.



Screenshots of the SHIVER webapp, developed and maintained by Ben Davison (U. Sheffield). Left: the site's landing page. Right: an example of am ice velocity time series generated for Whisky Glacier, showing the surge reported in Davison et al. (in press). SHIVER provides both point-and-click and programmatic access to a comprehensive collection of remotely sensed Antarctic and Greenland ice-surface velocity data, including the new SAR dataset described above (Davison et al., in prep.).

Conclusions

- Portions of Flask Glacier are free of firn.
- This allows an extensive surface drainage system to develop, including streams and lakes.
- A lake we instrumented drained nearly completely, with no surface outflow.
- Field data to be retrieved this coming season could reveal how lake drainage and other surface-to-bed meltwater flow affects ice flow.
- A survey of remotely sensed ice velocities has revealed a range of ice-dynamic changes with diverse causes on the Antarctic Peninsula.

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